

Rootkit

Adversaries may use rootkits to hide the presence of programs, files, network connections, services, drivers, and other system components. Rootkits are programs that hide the existence of malware by intercepting/hooking and modifying operating system API calls that supply system information. ^[1]

Rootkits or rootkit enabling functionality may reside at the user or kernel level in the operating system or lower, to include a hypervisor or [System Firmware](#). ^[2] Rootkits have been seen for Windows, Linux, and Mac OS X systems. ^[3] ^[4]

Rootkits that reside or modify boot sectors are known as [Bootkits](#) and specifically target the boot process of the operating system.

ID: T1014

Sub-techniques: No sub-techniques

①

Tactic: [Stealth](#)

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Platforms: Linux, Windows, macOS

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Procedure Examples

ID	Name	Description
G0007	APT28	APT28 has used a UEFI (Unified Extensible Firmware Interface) rootkit known as LoJax . ^{[5][6]}
G0096	APT41	APT41 deployed rootkits on Linux systems. ^{[7][8]}
C0046	ArcaneDoor	ArcaneDoor included hooking the <code>processHostScanReply()</code> function on victim Cisco ASA devices. ^[9]
S0484	Carberp	Carberp has used user mode rootkit techniques to remain hidden on the system. ^[10]
S0572	Caterpillar WebShell	Caterpillar WebShell has a module to use a rootkit on a system. ^[11]
S1105	COATHANGER	COATHANGER hooks or replaces multiple legitimate processes and other functions on victim devices. ^[12]
S0502	Drovorub	Drovorub has used a kernel module rootkit to hide processes, files, executables, and network artifacts from user space view. ^[13]
S0377	Ebury	Ebury acts as a user land rootkit using the SSH service. ^{[14][15]}
S0047	Hacking_Team UEFI Rootkit	Hacking_Team UEFI Rootkit is a UEFI BIOS rootkit developed by the company Hacking Team to persist remote access software on some targeted systems. ^[16]
S0394	HiddenWasp	HiddenWasp uses a rootkit to hook and implement functions on the system. ^[17]
S0135	HIDEDRV	HIDEDRV is a rootkit that hides certain operating system artifacts. ^[18]
S0009	Hikit	Hikit is a Rootkit that has been used by Axiom . ^[19] ^[20]
S0601	Hildegard	Hildegard has modified <code>/etc/ld.so.preload</code> to overwrite <code>readdir()</code> and <code>readdir64()</code> . ^[21]
S0040	HTRAN	HTRAN can install a rootkit to hide network connections from the host OS. ^[22]
S1186	Line Dancer	Line Dancer can hook both the crash dump process and the Authentication, Authorization, and Accounting (AAA) functions on compromised machines to evade forensic analysis and authentication mechanisms. ^[9]
S0397	LoJax	LoJax is a UEFI BIOS rootkit deployed to persist remote access software on some targeted systems. ^[6]
S1220	MEDUSA	MEDUSA is a rootkit with command execution and credential logging capabilities. ^[23]
S0012	PoisonIvy	PoisonIvy starts a rootkit from a malicious file dropped to disk. ^[24]
S0458	Ramsay	Ramsay has included a rootkit to evade defenses. ^[25]
C0056	RedPenguin	During RedPenguin , UNC3886 used rootkits such as REPTILE and MEDUSA . ^[26]
S1219	REPTILE	REPTILE has the ability to hook kernel functions and modify functions data to achieve rootkit functionality such as hiding processes and network connections. ^[23]
G0106	Rocke	Rocke has modified <code>/etc/ld.so.preload</code> to hook <code>libc</code> functions in order to hide the installed dropper and mining software in process lists. ^[27]
S0468	Skidmap	Skidmap is a kernel-mode rootkit that has the ability to hook system calls to hide specific files and fake network and CPU-related statistics to make the CPU load of the infected machine always appear low. ^[28]

ID	Name	Description
S0221	Umbreon	Umbreon hides from defenders by hooking libc function calls, hiding artifacts that would reveal its presence, such as the user account it creates to provide access and undermining strace, a tool often used to identify malware. ^[32]
G1048	UNC3886	UNC3886 has used the publicly available rootkits REPTILE and MEDUSA on targeted VMs. ^[23]
S0022	Uroburos	Uroburos can use its kernel module to prevent its host components from being listed by the targeted system's OS and to mediate requests between user mode and concealed components. ^{[33][34]}
S0670	WarzoneRAT	WarzoneRAT can include a rootkit to hide processes, files, and startup. ^[35]
S0430	Winnti for Linux	Winnti for Linux has used a modified copy of the open-source userland rootkit Azazel, named libxslinux.so, to hide the malware's operations and network activity. ^[36]
G0044	Winnti Group	Winnti Group used a rootkit to modify typical server functionality. ^[37]
S0027	Zeroaccess	Zeroaccess is a kernel-mode rootkit. ^[38]

Mitigations

This type of attack technique cannot be easily mitigated with preventive controls since it is based on the abuse of system features.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0377	Detection of Kernel/User-Level Rootkit Behavior Across Platforms	AN1061	Unauthorized or anomalous loading of kernel-mode drivers or DLLs, concealed services, or abnormal modification of boot components indicative of rootkit activity.
		AN1062	Abnormal loading of kernel modules, direct tampering with /dev, /proc, or LD_PRELOAD behaviors hiding processes or files.
		AN1063	Execution of unsigned kernel extensions (KEXTs), tampering with LaunchDaemons, or userspace hooks into system libraries.

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